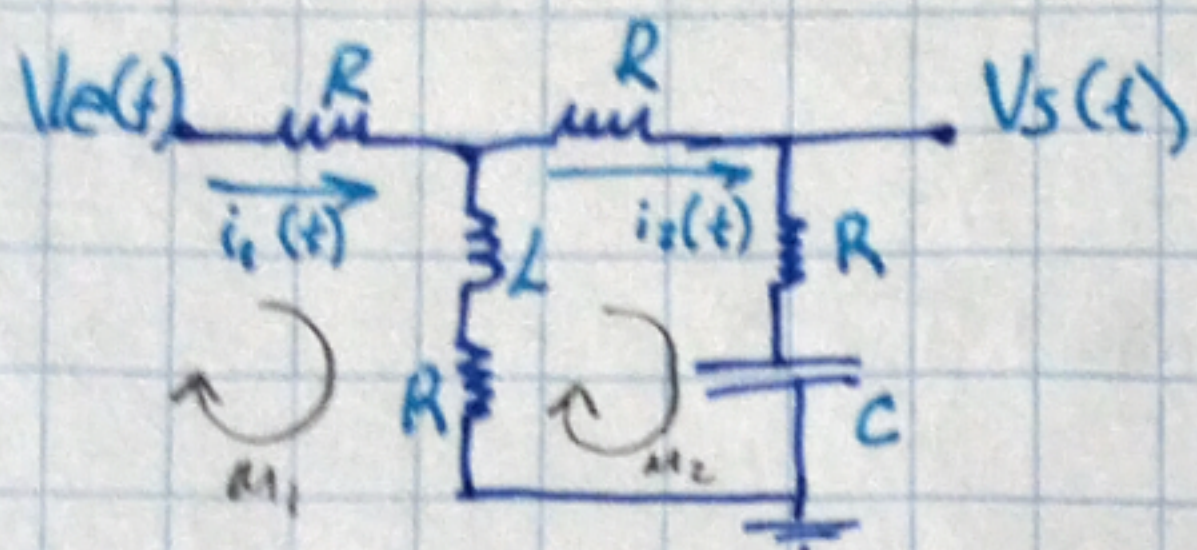


Práctica 1. Diseño de Controlador para un Sistema de Segundo Orden.



Nota: Función de Transferencia depende de los componentes RLC.

→ Ecuaciones Principales.

- $V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$ Ec. Malla 1
- $L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$ Ec. Malla 2
- $V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$

→ Transformada de Laplace.

- $V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$
- $(Ls + R) I_1(s) = (Ls + \frac{1}{Cs} + 3R) I_2(s)$
- $I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$
- $V_s(s) = (R + \frac{1}{Cs}) I_2(s)$

→ Procedimiento Algebraico.

$$\begin{aligned}
 \bullet I_1(s) &= \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s) = \left\{ \frac{1}{Ls + R} \left[(Ls + \frac{1}{Cs} + 3R) \frac{Cs}{Cs} \right] \right\} I_2(s) \\
 &= \frac{1}{Ls + R} \left(\frac{CLs^2 + 1 + 3CRs}{Cs} \right) I_2(s) = \left(\frac{CLs^2 + 1 + 3CRs}{(Cs)(Ls + R)} \right) I_2(s) \\
 I_1(s) &= \left(\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} \right) I_2(s)
 \end{aligned}$$

$$\begin{aligned} \bullet V_s(s) &= R I_2(s) + \frac{I_2(s)}{Cs} = \left(R + \frac{1}{Cs}\right) I_2(s) = \left[\left(R + \frac{1}{Cs}\right) \frac{Cs}{Cs}\right] I_2(s) \\ &= \left(\frac{CRs + 1}{Cs}\right) I_2(s) \end{aligned}$$

$$\begin{aligned} \bullet V_e(s) &= (Ls + 2R) I_2(s) - (Ls + R) I_2(s) \\ &= (Ls + 2R) \left(\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs}\right) I_2(s) - (Ls + R) I_2(s) \\ &= \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} - (Ls + R)\right] I_2(s) \\ &= \left[\frac{CL^2s^3 + 3CLR s^2 + Ls + 2CLR s^2 + 6CR^2s + 2R}{CLs^2 + CRs} - (Ls + R)\right] I_2(s) \\ &= \left(\frac{CL^2s^3 + 5CLR s^2 + Ls + 6CR^2s + 2R - Ls(CLs^2 + CRs) - R(CLs^2 + CRs)}{CLs^2 + CRs}\right) I_2(s) \\ &= \left(\frac{CL^2s^3 + 5CLR s^2 + Ls + 6CR^2s + 2R - CL^2s^3 - CLR s^2 - CLR s^2 - CR^2s}{CLs^2 + CRs}\right) I_2(s) \\ &= \left(\frac{3CLR s^2 + Ls + 5CR^2s + 2R}{CLs^2 + CRs}\right) I_2(s) \\ &= \frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s) \end{aligned}$$

→ Función de Transferencia.

$$\begin{aligned} \frac{V_s(s)}{V_e(s)} &= \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)} = \frac{(CRs + 1)(CLs^2 + CRs)}{Cs(3CLR s^2 + 5CR^2s + Ls + 2R)} \\ &= \frac{(CRs + 1) \cancel{Cs} (Ls + R)}{\cancel{Cs} (3CLR s^2 + 5CR^2s + Ls + 2R)} = \frac{(CRs + 1)(Ls + R)}{3CLR s^2 + 5CR^2s + Ls + 2R} \\ &= \frac{CLRs^2 + CR^2s + Ls + R}{3CLR s^2 + 5CR^2s + Ls + 2R} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \end{aligned}$$

→ Error en Estado Estacionario.

$$e(s) = \lim_{s \rightarrow 0} s \cdot V_e(s) \left(1 - \frac{V_s(s)}{V_e(s)}\right)$$

$$\begin{aligned} e(s) &= s \cdot \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R}\right) \\ &= \frac{R}{2R} \quad \left| e(t) = \frac{1}{2}V ; \text{error de } 0.5V \right. \end{aligned}$$

$$V_e(t) = \begin{cases} 0V, & t \in (-\infty, 0) \\ 1V, & t \in [0, \infty) \end{cases}$$

$V_e(s) = 1/s$

→ Error en Estudio Estacionario

• Estabilidad en lazo abierto. $R = 4.7 \text{ k}\Omega$ $L = 10 \text{ mH}$ $C = 330 \text{ nF}$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(5CR^2 + L) \pm \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)}$$

$$b = 5CR^2 + L = [5(330 \times 10^{-6})(4.7 \times 10^3)^2] + 10 \times 10^{-3} = 36448.51$$

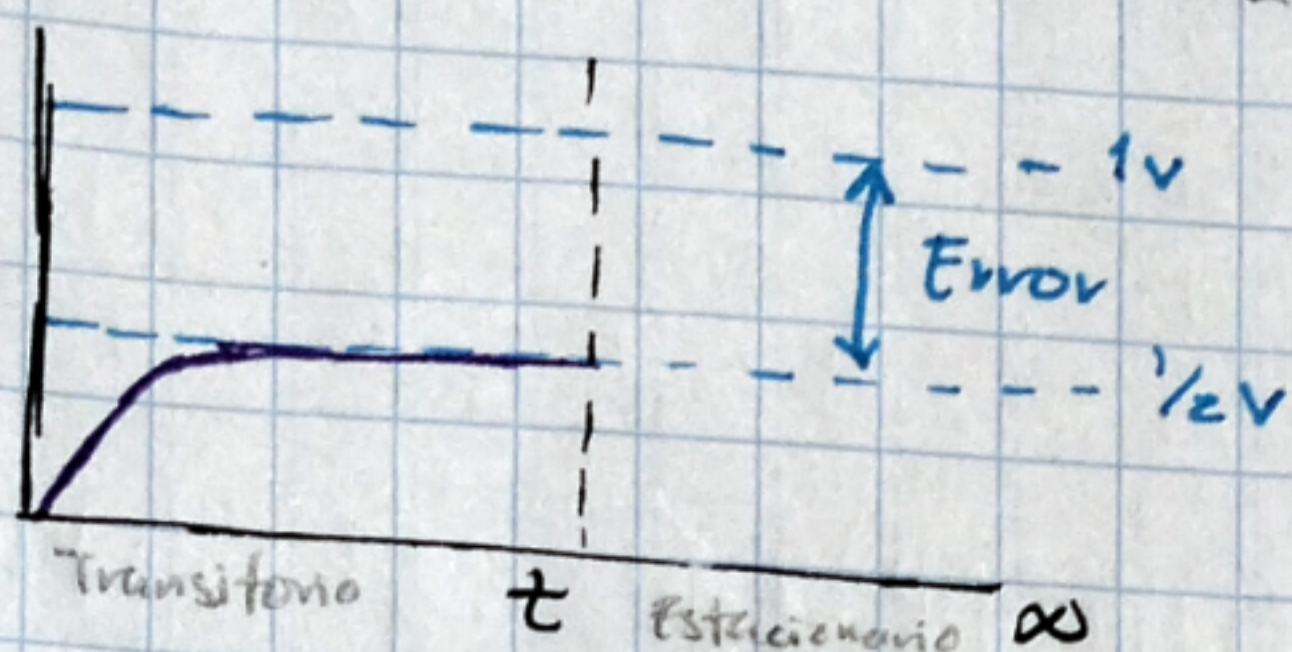
$$c = 2R = 2(4.7 \times 10^3) = 9.4 \times 10^3$$

$$a = 3CLR = 3(330 \times 10^{-6})(10 \times 10^{-3})(4.7 \times 10^3) = 0.04653$$

$$\lambda_{1,2} = \frac{-(36448.51) \pm \sqrt{(36448.51)^2 - 4(0.04653)(9400)}}{2(0.04653)}$$

$$\lambda_1 = -0.257898$$

$$\lambda_2 = -783333.29035 = -7.8333 \times 10^5$$



• La respuesta es sobrecorregida.

→ Modelo de Ecuaciones Integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir las corrientes:

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$\hookrightarrow i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \cdot \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$